

Semester - III			
Sl. No.	Course Code	Course Title	Page No.
1	MA231TC	Linear Algebra and Probability Theory	1
2	CV232TA / ME232TB / BT232TD	Environment and Sustainability / Material Science for Engineers / Bioinspired Engineering	3 / 5 / 7
3	CS233AI	Data Structures and Applications	9
4	CS234AI	Applied Digital Logic Design and Computer Organization	11
5	CS235AT	Operating Systems	13
6	CS236AT	Quantum Computing Essentials	15
7	CS238DL	Applied Design thinking lab	17
8	MAT149AT	Bridge Course Mathematics	18

Semester - IV			
Sl. No.	Course Code	Course Title	Page No.
1	CS241AT	Discrete mathematical structures and combinatorics	19
2	CV232TA / ME232TB / BT232TD	Environment and Sustainability / Material Science for Engineers / Bioinspired Engineering	3 / 5 / 7
3	CD343AI	Design and Analysis of Algorithms	21
4	CS345AI	Embedded Systems, IoT and Edge AI	23
5	CS246AT	Computer Networks	25
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SKILL LABS			
Sl.	Course Code	Course Title	Page No.
1	SLCSP31A	Domain Engineering Skill Lab	42
2	SLCSP31B	Domain Engineering Skill Lab	43



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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2025 Scheme template – UG Programs														
B.E in Electronics and Communication Engineering (ECE)														
III SEMESTER														
Sl.	Course Code	Course Title	BoS	Credit Allocation				Category	CIE Duration (Hrs)	Max Marks CIE		SEE Duration (Hrs)	Max Marks SEE	
				L	T	P	Total			Theory	Lab		Theory	Lab
1	MA231TC	Linear Algebra and Probability Theory	MA	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
2	CV232TA / ME232TB / BT232TD	Environment and Sustainability / Material Science for Engineers / Bioinspired Engineering	CV / ME / BT	1	0	0	1	Theory	1	80	***	2	100	***
3	CS233AI	Data Structures and Applications	CS	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
4	CS234AI	Applied Digital Logic Design and Computer Organization	CS	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
5	CS235AT	Operating Systems	CS	3	0	0	3	Theory	2	100	***	3	100	***
6	CS236AT	Quantum Computing Essentials	CS	3	0	0	3	Theory	2	100	***	3	100	***
7	CS238DL	Applied Design thinking lab	CS	0	0	1	1	Lab	2	50	50	3	50	50
8	MAT149AT	Bridge Course Mathematics	MA	3	0	0	0 [†]	Theory	2	100	***	—	***	***
Total				19	0	4	20							

[†] Audit

L duration: 15 hrs/credit	T duration: 30 hrs/credit	P duration: 30 hrs/credit	S duration: Total credits * hrs/sem - (L + T + P) durations
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Important Note: It is mandatory for all students to complete the Domain Engineering Skill Lab (SLCSP31A & SLCSP31B) in both III and IV Semesters.



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2025 Scheme template – UG Programs														
B.E in Electronics and Communication Engineering (ECE)														
IV SEMESTER														
Sl.	Course Code	Course Title	BoS	Credit Allocation				Category	CIE Duration (Hrs)	Max Marks CIE		SEE Duration (Hrs)	Max Marks SEE	
				L	T	P	Total			Theory	Lab		Theory	Lab
1	CS241AT	Discrete mathematical structures and combinatorics	CS	3	0	0	3	Theory	2	100	***	3	100	***
2	CV232TA / ME232TB / BT232TD	Environment and Sustainability / Material Science for Engineers / Bioinspired Engineering	CV / ME / BT	3	0	0	3	Theory	2	100	***	3	100	***
3	CD343AI	Design and Analysis of Algorithms	CD	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
4	CS345AI	Embedded Systems, IoT and Edge AI	CS	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
5	CS246AT	Computer Networks	CS	3	1	0	4	Theory	2	100	***	3	100	***
6	CS248ATX	Professional Electives	CS	3	0	0	3	Theory	2	100	***	3	100	***
7	HS248LX	AE Course	HS	0	0	1	1	Lab	2	50	50	3	50	50
8	CS139AT	Bridge Course C Programming	CS	3	0	0	0 [†]	Theory	2	100	***	—	***	***
Total				21	1	3	22							

[†] Audit

L duration: 15 hrs/credit	T duration: 30 hrs/credit	P duration: 30 hrs/credit	S duration: Total credits * hrs/sem - (L + T + P) durations
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Important Note: It is mandatory for all students to complete the Domain Engineering Skill Lab (SLCSP31A & SLCSP31B) in both III and IV Semesters.

Semester - IV			
Professional Electives			
GROUP-A			
Sl. No.	Course Code	Course Title	Page No.
1	CS248ATA	Graph Theory	??
2	CS248ATB	Computer Graphics and Extended Reality	??
3	CI248ATC	Quantum Algorithms and Key Distribution	??
4	CS248ATD	Natural Language Processing	??
5	CS355TBF	Linux Internals	??
6	CS365TDB	Operation Research	??

Semester - IV			
AE Courses			
Sl. No.	Course Code	Course Title	Page No.
1	HS238LA	National Service Scheme	??
2	HS238LB	National Cadet Corps	??



Semester - III					
Linear Algebra and Probability Theory Category: Professional Core Course (Common to CS and CY) (Theory & Practical)					
Course Code	:	MA231TC	CIE	:	100 + 50 Marks
Credit distribution L:T:P	:	3:0:1	SEE	:	100 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45	CIE Duration	:	1.5 Hours
Total credits:	:	4	SEE Duration	:	3 Hours
??					
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (20 Marks), Lab test (10 Marks) and Innovative experiment/Concept Design and Implementation (20 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE REDUCED TO 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B		



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Question 3 or 4	18
5 & 6	Unit - III: Question 5 or 6	18
7 & 9	Unit - IV: Question 7 or 8	18
9 & 10	Unit - V: Question 9 or 10	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Write Up	10
2	Conduction of the Experiments	20
3	Viva	20
TOTAL		50



Semester - III					
Environment and Sustainability Category: Professional Core Course (Common to CS,CD,CY and CI Programs) (Theory)					
Course Code	:	CV232TA	CIE	:	80 Marks
Credit distribution L:T:P	:	1:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	15:0:0:15	CIE Duration	:	1 Hours
Total credits:	:	1	SEE Duration	:	2 Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Compulsory	18
5 & 6	Unit - III: Compulsory	18
7 & 9	Unit - IV: Compulsory	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
9 & 10	Unit - V: Compulsory	18
TOTAL		100



Semester - III					
Material Science for Engineers Category: Professional Core Course (Common to CS,CD,CY and CI Programs) (Theory)					
Course Code	:	ME232TB	CIE	:	80 Marks
Credit distribution L:T:P	:	1:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	15:0:0:15	CIE Duration	:	1 Hours
Total credits:	:	1	SEE Duration	:	2 Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Compulsory	18
5 & 6	Unit - III: Compulsory	18
7 & 9	Unit - IV: Compulsory	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
9 & 10	Unit - V: Compulsory	18
TOTAL		100



Semester - III					
Bioinspired Engineering Category: Professional Core Course (Common to CS,CD,CY and CI Programs) (Theory)					
Course Code	:	BT232TD	CIE	:	80 Marks
Credit distribution L:T:P	:	1:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	15:0:0:15	CIE Duration	:	1 Hours
Total credits:	:	1	SEE Duration	:	2 Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Compulsory	18
5 & 6	Unit - III: Compulsory	18
7 & 9	Unit - IV: Compulsory	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
9 & 10	Unit - V: Compulsory	18
TOTAL		100



Semester - III					
Data Structures and Applications Category: Professional Core Course (Common to CS and CY) (Theory & Practical)					
Course Code	:	CS233AI	CIE	:	100 + 50 Marks
Credit distribution L:T:P	:	3:0:1	SEE	:	100 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45	CIE Duration	:	1.5 Hours
Total credits:	:	4	SEE Duration	:	3 Hours
??					
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (20 Marks), Lab test (10 Marks) and Innovative experiment/Concept Design and Implementation (20 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE REDUCED TO 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B		



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Question 3 or 4	18
5 & 6	Unit - III: Question 5 or 6	18
7 & 9	Unit - IV: Question 7 or 8	18
9 & 10	Unit - V: Question 9 or 10	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Write Up	10
2	Conduction of the Experiments	20
3	Viva	20
TOTAL		50



Semester - III					
Applied Digital Logic Design and Computer Organization					
Category: Professional Core Course					
(Common to CS, CD and CY)					
(Theory & Practical)					
Course Code	:	CS234AI	CIE	:	100 + 50 Marks
Credit distribution L:T:P	:	3:0:1	SEE	:	100 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45	CIE Duration	:	1.5 Hours
Total credits:	:	4	SEE Duration	:	3 Hours
??					
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (20 Marks), Lab test (10 Marks) and Innovative experiment/Concept Design and Implementation (20 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE REDUCED TO 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B		



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Question 3 or 4	18
5 & 6	Unit - III: Question 5 or 6	18
7 & 9	Unit - IV: Question 7 or 8	18
9 & 10	Unit - V: Question 9 or 10	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Write Up	10
2	Conduction of the Experiments	20
3	Viva	20
TOTAL		50



Semester - III					
Operating Systems Category: Professional Core Course (Common to CS, CD and CI) (Theory)					
Course Code	:	CS235AT	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - III					
Quantum Computing Essentials Category: Professional Core Course (Common to CS, CD and CI) (Theory)					
Course Code	:	CS236AT	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - III					
Applied Design thinking lab Category: Professional Core Course (Common to CS,CD,CY and CI Programs) (Practical)					
Course Code	:	CS238DL	CIE	:	50 + 50 Marks
Credit distribution L:T:P	:	0:0:1	SEE	:	50 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	0:0:30:0	CIE Duration	:	2 Hours
Total credits:	:	1	SEE Duration	:	3 Hours
??					
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 20 MARKS.	20
MAXIMUM MARKS FOR THE CIE		80

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Objective type questions covering entire syllabus	20
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
TOTAL		52



Semester - III					
Bridge Course Mathematics Category: Bridge Course (Common to CS,CD,CY and CI Programs) (Theory)					
Course Code	:	MAT149AT	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	— Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	— Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



Semester - IV					
Discrete mathematical structures and combinatorics Category: Professional Core Course (Common to Common to CS and CY Programs) (Theory)					
Course Code	:	CS241AT	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV					
Design and Analysis of Algorithms Category: Professional Core Course (Common to CS,CD,CY and CI Programs) (Theory & Practical)					
Course Code	:	CD343AI	CIE	:	100 + 50 Marks
Credit distribution L:T:P	:	3:0:1	SEE	:	100 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45	CIE Duration	:	1.5 Hours
Total credits:	:	4	SEE Duration	:	3 Hours
??					
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (20 Marks), Lab test (10 Marks) and Innovative experiment/Concept Design and Implementation (20 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE REDUCED TO 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B		



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Question 3 or 4	18
5 & 6	Unit - III: Question 5 or 6	18
7 & 9	Unit - IV: Question 7 or 8	18
9 & 10	Unit - V: Question 9 or 10	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Write Up	10
2	Conduction of the Experiments	20
3	Viva	20
TOTAL		50



Semester - IV					
Embedded Systems, IoT and Edge AI Category: Professional Core Course (Common to CS and CI Programs) (Theory & Practical)					
Course Code	:	CS345AI	CIE	:	100 + 50 Marks
Credit distribution L:T:P	:	3:0:1	SEE	:	100 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45	CIE Duration	:	1.5 Hours
Total credits:	:	4	SEE Duration	:	3 Hours
??					
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (20 Marks), Lab test (10 Marks) and Innovative experiment/Concept Design and Implementation (20 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE REDUCED TO 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B		



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Question 3 or 4	18
5 & 6	Unit - III: Question 5 or 6	18
7 & 9	Unit - IV: Question 7 or 8	18
9 & 10	Unit - V: Question 9 or 10	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Write Up	10
2	Conduction of the Experiments	20
3	Viva	20
TOTAL		50



Semester - IV					
Computer Networks					
Category: Professional Core Course					
(Common to CS,CD,CY and CI Programs)					
(Theory & Tutorial)					
Course Code	:	CS246AT	CIE	:	100 Marks
Credit distribution L:T:P	:	3:1:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:30:0:45	CIE Duration	:	2 Hours
Total credits:	:	4	SEE Duration	:	3 Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY & TUTORIAL)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B		
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16
9 & 10	Unit - V: Compulsory	16



RUBRICS FOR SEMESTER END EXAMINATION (THEORY & TUTORIAL) (Continued)		
Q.NO.	CONTENTS	MARKS
	TOTAL	100



Semester - IV					
Graph Theory Category: Professional Core Elective (Only CS) (Theory)					
Course Code	:	CS248ATA	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B		
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV					
Computer Graphics and Extended Reality Category: Professional Core Elective (Only CS) (Theory)					
Course Code	:	CS248ATB	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV					
Quantum Algorithms and Key Distribution Category: Professional Core Elective (Common to CS, CI and CD Programs) (Theory)					
Course Code	:	CI248ATC	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV					
Natural Language Processing Category: Professional Core Elective (Only CS) (Theory)					
Course Code	:	CS248ATD	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV					
Linux Internals					
Category: Professional Core Elective (Common to CS, CI, CY and CD Programs) (Theory)					
Course Code	:	CS355TBF	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV					
Operation Research Category: Professional Core Elective (Only CS) (Theory)					
Course Code	:	CS365TDB	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B		
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
5 & 6	Unit - III: Compulsory	16
7 & 9	Unit - IV: Compulsory	16



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
9 & 10	Unit - V: Compulsory	16
TOTAL		100



Semester - IV					
National Service Scheme Ability Enhancement Course (All programs) (Practical)					
Course Code	:	HS238LA	CIE	:	50 + 50 Marks
Credit distribution L:T:P	:	0:0:1	SEE	:	50 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	0:0:30:0	CIE Duration	:	2 Hours
Total credits:	:	1	SEE Duration	:	3 Hours
??					
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 20 MARKS.	20
MAXIMUM MARKS FOR THE CIE		80

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Objective type questions covering entire syllabus	20
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
TOTAL		52



Semester - IV					
National Cadet Corps Ability Enhancement Course (All programs) (Practical)					
Course Code	:	HS238LB	CIE	:	50 + 50 Marks
Credit distribution L:T:P	:	0:0:1	SEE	:	50 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	0:0:30:0	CIE Duration	:	2 Hours
Total credits:	:	1	SEE Duration	:	3 Hours
??					
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 20 MARKS.	20
MAXIMUM MARKS FOR THE CIE		80

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Objective type questions covering entire syllabus	20
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
TOTAL		52



Semester - IV					
Bridge Course C Programming Category: Bridge Course (Common to Common to CS,CD,CY and CI Programs) (Theory)					
Course Code	:	CS139AT	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	— Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	— Hours
Course Outcomes: After completing the course, the students will be able to:					
1. ??					
References					
[1] ??					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



Domain Engineering Skill Lab Category: Skill Lab (Common to CS,CD,CY and CI Programs) (Practical)					
Course Code	:	SLCSP31A	CIE	:	50 + 50 Marks
Credit distribution L:T:P	:	0:0:1	SEE	:	50 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	0:0:30:0	CIE Duration	:	2 Hours
Total credits:	:	1	SEE Duration	:	3 Hours
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] Razavi2000					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 20 MARKS.	20
MAXIMUM MARKS FOR THE CIE		80

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Objective type questions covering entire syllabus	20
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
TOTAL		52



Domain Engineering Skill Lab Category: Skill Lab (Common to CS,CD,CY and CI Programs) (Practical)					
Course Code	:	SLCSP31B	CIE	:	50 + 50 Marks
Credit distribution L:T:P	:	0:0:1	SEE	:	50 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	0:0:30:0	CIE Duration	:	2 Hours
Total credits:	:	1	SEE Duration	:	3 Hours
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] Razavi2000					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 20 MARKS.	20
MAXIMUM MARKS FOR THE CIE		80

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Objective type questions covering entire syllabus	20
2	Unit - I: Compulsory	16
3 & 4	Unit - II: Compulsory	16
TOTAL		52